

RRC-Raptor-V4H SCIF/HSCIF

- User's Manual: Software

V0.0.3

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i. VERSION LIST

Description	Version	Date	Author
RTX_SCIF/HSCIF	V0.0.1	2024/07/22	Jerry
RTX_SCIF/HSCIF	V0.0.2	2024/10/29	Jerry
Translate the content into English.	V0.0.3	2024/11/20	Jerry

ii. HASHTAG

#V4H

#UART

#SCIF

#HSCIF

#BSP

1. Overview

1.1. Overview

Target : This manual explains the driver module that controls the SCIF and HSCIF on R-Car.

1.2. Function

This module controls the Serial Communication Interface with FIFO (SCIF) on R-Car . This module supports the following functions.

- Data transmission and reception for RS232C
- Control of communication settings
 - Baud rate 9600, 19200, 38400, 57600, 115200[bps]
 - Parity bit (none/ODD/EVEN)
 - Stop bit (1bit or 2bit)
 - Data transfer bit length (7bit or 8bit)Each setting can be changed from the standard ioctl for tty.
- Hardware Flow Control
 - Support: SCIF0, SCIF1, SCIF3, SCIF4
 - Not support: SCIF2, SCIF5

This module controls the High Speed Serial Communication Interface with FIFO (HSCIF) on R-Car. This module supports the following functions.

- Data transmission and reception for RS232C
- Control of communication settings
 - Baud rate 9600, 19200, 38400, 57600, 115200, 230400, 460800, 921600, and 1000000 ~ 3000000(Max)[bps]
 - Parity bit (none/ODD/EVEN)
 - Stop bit (1bit or 2bit)
 - Data transfer bit length (7bit or 8bit)Each setting can be changed from the standard ioctl for tty.
- Hardware Flow Control

2. Directory Configuration

The directory configuration is shown below.

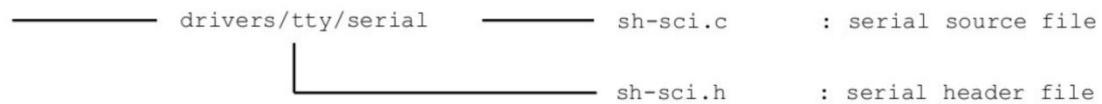


Figure 1. Directory Configuration

3. Enable Hardware Flow Control

The following description explain how to enable hardware flow control function. Add "uart-has-rtsts" property to use the hardware flow control function. The following description indicates a difference from the default setting. ("+": Setting after the modification)

```
&hscif3 {
    pinctrl-0 = <&hscif3_pins>;
    pinctrl-names = "default";
    uart-has-rtsts;
    /* Please use exclusively to the scif0 node */
    status = "okay";
};
```

Figure 2. Enable flow control

4. Kernel Configuration Setting

To enable the functions of this module, make the following setting with Kernel Configuration.

```
General setup  --->
  [ ] Embedded system *1

Device Drivers  --->
  Character devices  --->
    Serial drivers  --->
      <*> SuperH SCI(F) serial port support
      (11) Maximum number of SCI(F) serial ports *2
      [*] Support for console on SuperH SCI(F) *2
      [*] Support for early console on SuperH SCI(F) *2
      [*] DMA support *2 *3
```

Figure 3 Configuration Setting

5. Connect

Provide a UART interface so that users can add their own UART devices.

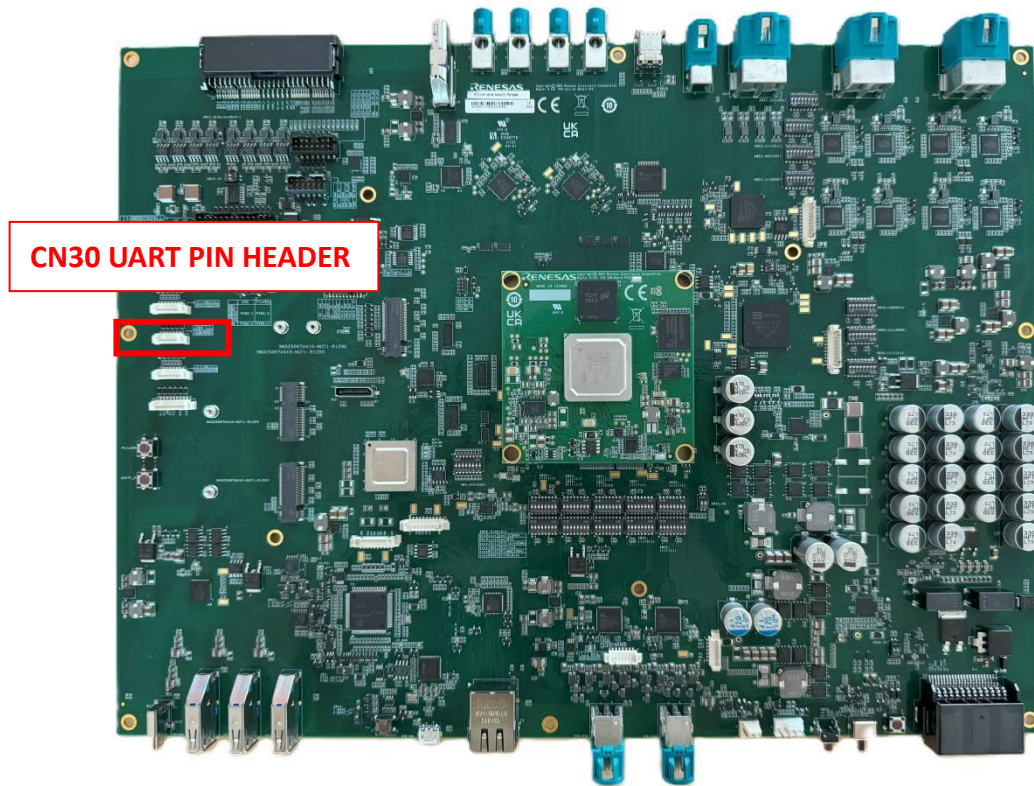


Figure 4 Connect

CN30 UART PIN HEADER

PIN	SYMBOL
1	D3.3V_SPI
2	GP1_27/HSCIF3_HCTS3N_V (CTS)
3	GP1_26/HSCIF3_HRTS3N_V (RTS)
4	GND
5	GP1_24/SL_SW1_V (RX)
6	GP1_28/HSCIF3_HTX3_V (TX)

6. UART Tool

Use stty can set UART characteristics .

stty -F /dev/ttySC3 115200 cs8 -cstopb -parenb

//Set the baudrate 115200, 8 bits, 1 stop bis, no parity

stty -F /dev/ttySC3 -crtcts

//Stop flow control

stty -F /dev/ttySC3 crtscts

//Set flow control

7. APPENDIX